

Exercises

Adding, changing and deleting data

Use webshop.db for all exercises in this part. Before you start, check that the orders are there: `SELECT COUNT(*) FROM orders;` should give 15. If not, run `webshop_reset.sql` and press Ctrl+S. When you are done, click Revert Changes so your data is the same as everyone else's.

Exercise 1

Add yourself as customer number 11 (use a made-up email).

Exercise 2

Add two new products in one INSERT: Scarf (Accessories, 229 kr, 15 in stock) and Gloves (Accessories, 199 kr, 20 in stock).

Exercise 3

Customer 7 (Emma) orders 2 Beanies (product 10, 179 kr). Make the order (order 16) and the order item.

Exercise 4

Try to add an order item with quantity 0. Which rule stops you?

Exercise 5

Order 12 has been shipped. Change its status.

Exercise 6

The Water Bottle (product 5) is back in stock: 50 pieces.

Exercise 7

Raise the price of all Accessories by 10%.

Exercise 8

Delete the cancelled order. Watch out: its items must go first! Why?

Exercise 9

Finally: click Revert Changes so your data matches the teacher's again. Check: 15 orders?

Designing a good database

Exercises 10 to 15 are done on paper or in a drawing tool. You don't need a database for them. Exercise 16 is the only one where you write SQL, and it uses a new database file, not `webshop.db`.

Exercise 10

Look at this table: student | phone_numbers | course1 | course2 | course3. List every problem you can find.

Exercise 11

In order_sheet, which normal form does the products column break? How would you fix it?

Exercise 12

A table has: order_id | customer_id | customer_email | order_date. Which column is in the wrong place? Why?

Exercise 13

Music school (in pairs): 'Students take lessons from teachers. A lesson has a date, time, room and instrument. One teacher can teach many instruments.' Underline the things.

Exercise 14

Find the relationships in the music school. Which are 1:N and which are N:M?

Exercise 15

Draw the ER diagram for the music school.

Exercise 16

Write the CREATE TABLE statements for the music school in a new file music.db.

Database: create a new file. Click New Database, name it music.db, and click Cancel in the 'Edit table definition' window. Don't create these tables in webshop.db.